

E-commerce Business Performance & Sales Analysis

SQL Project Using MYSQL

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-- KEY METRICS — Headline KPIs

-- =====

-- TOP 5 Country YTD Total Sales IN (2023)

SELECT

country,

year(o.order_date) Year,

round(sum(oi.quantity * oi.unit_price),1) AS Total_Sales_In_Million

FROM orders o

LEFT JOIN order_items oi

ON o.order_id = oi.order_id

LEFT JOIN Customers c

on o.customer_id = c.customer_id

Where year(o.order_date)= 2023

GROUP BY YEAR(order_date),country

ORDER BY Total_Sales_In_Million DESC LIMIT 5;

Result Grid			
	country	Year	Total_Sales
▶	USA	2023	411594.4
	Japan	2023	325754.4
	India	2023	298307.4
	Germany	2023	277835.9
	UK	2023	268935.5

-- TOP 5 Country YTD Total Sales IN (2024)

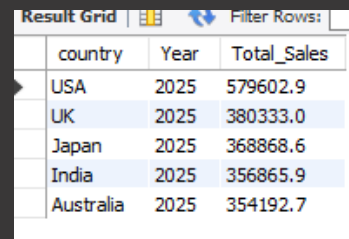
```
SELECT
    country,
    year(o.order_date) Year,
    round(sum(oi.quantity * oi.unit_price),1) AS Total_Sales_In_Million
FROM orders o
LEFT JOIN order_items oi
ON o.order_id = oi.order_id
LEFT JOIN Customers c
on o.customer_id = c.customer_id
Where year(o.order_date)= 2024
GROUP BY YEAR(order_date),country
ORDER BY Total_Sales_In_Million DESC LIMIT 5;
```



	country	Year	Total_Sales
▶	USA	2024	605149.6
	India	2024	421235.4
	Canada	2024	406110.7
	Australia	2024	386457.2
	UK	2024	385328.4

-- TOP 5 Country YTD Total Sales IN (2025)

```
SELECT
    country,
    year(o.order_date) Year,
    round(sum(oi.quantity * oi.unit_price),1) AS Total_Sales_In_Million
FROM orders o
LEFT JOIN order_items oi
ON o.order_id = oi.order_id
LEFT JOIN Customers c
on o.customer_id = c.customer_id
Where year(o.order_date)= 2025
GROUP BY YEAR(order_date),country
ORDER BY Total_Sales_In_Million DESC LIMIT 5;
```



	country	Year	Total_Sales
▶	USA	2025	579602.9
	UK	2025	380333.0
	Japan	2025	368868.6
	India	2025	356865.9
	Australia	2025	354192.7

-- TOP 5 City YTD Total Sales IN (2023)

```
SELECT
    city,
    year(o.order_date) Year,
    round(sum(oi.quantity * oi.unit_price),1) AS Total_Sales_In_Million
FROM orders o
LEFT JOIN order_items oi
```

```

ON o.order_id = oi.order_id
LEFT JOIN Customers c
on o.customer_id = c.customer_id
Where year(o.order_date)= 2023
GROUP BY YEAR(order_date), City
ORDER BY Total_Sales_In_Million DESC LIMIT 5;

```

Result Grid			
	city	Year	Total_Sales
▶	Munich	2023	178137.4
	Paris	2023	177306.5
	Osaka	2023	168332.3
	London	2023	167567.3
	Tokyo	2023	157422.1

```

-- TOP 5 City YTD Total Sales IN (2024)
SELECT
    city,
    year(o.order_date) Year,
    round(sum(oi.quantity * oi.unit_price),1) AS Total_Sales_In_Million
FROM orders o
LEFT JOIN order_items oi
ON o.order_id = oi.order_id
LEFT JOIN Customers c
on o.customer_id = c.customer_id
Where year(o.order_date)= 2024
GROUP BY YEAR(order_date), City
ORDER BY Total_Sales_In_Million DESC LIMIT 5;

```

Result Grid			
	city	Year	Total_Sales
▶	Singapore	2024	242734.9
	Chicago	2024	220839.5
	Mumbai	2024	218658.6
	Sydney	2024	217929.4
	Vancouver	2024	217366.0

```

-- TOP 5 City YTD Total Sales IN (2025)
SELECT
    city,
    year(o.order_date) Year,
    round(sum(oi.quantity * oi.unit_price),1) AS Total_Sales_In_Million
FROM orders o
LEFT JOIN order_items oi
ON o.order_id = oi.order_id
LEFT JOIN Customers c
on o.customer_id = c.customer_id
Where year(o.order_date)= 2025
GROUP BY YEAR(order_date), City
ORDER BY Total_Sales_In_Million DESC LIMIT 5;

```

Result Grid			
	city	Year	Total_Sales
▶	Chicago	2025	247427.6
	Paris	2025	215729.5
	London	2025	215167.4
	Tokyo	2025	196594.0
	Dubai	2025	194044.9

-- Total Sales

Select

```
round(sum(quantity * unit_price)/1000000,3) as Total_Sales_In_Million
```

```
FROM order_items;
```

Total_Sales_In_Million
9.395

```
-- Total Orders
```

```
SELECT
```

```
count(order_id)/1000 AS Total_Orders_In_Thousand
```

```
FROM orders;
```

Total_Orders_In_Thousand
10.0000

```
-- Average Order Value KPI
```

```
SELECT
```

```
round(sum(quantity * unit_price) /(count(distinct o.order_id)),2) as AOV
```

```
FROM orders o
```

```
left join order_items oi
```

```
on o.order_id =oi.order_id;
```

AOV
939.52

```
-- Total Qty Sold KPI
```

```
SELECT
```

```
SUM(quantity) as total_qty_sold
```

```
from order_items;
```

total_qty_sold
43895

-- Total Profit KPI

SELECT

SUM(quantity * unit_price) as total_sales,

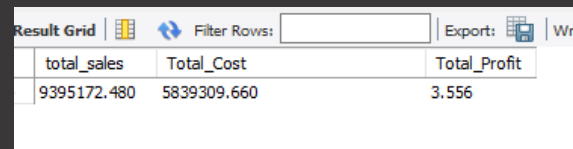
SUM(quantity * Cost_Price) as Total_Cost,

round((SUM(quantity * unit_price)- SUM(quantity * Cost_Price))/1000000,3) as Total_Profit

FROM order_items oi

LEFT JOIN products p

on oi.product_id = p.product_id;



total_sales	Total_Cost	Total_Profit
9395172.480	5839309.660	3.556

-- Total Product Cost KPI

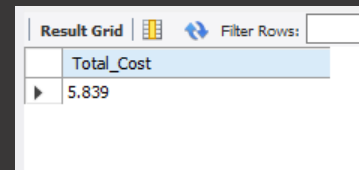
SELECT

round(SUM(quantity * Cost_Price)/1000000,3) AS Total_Cost

FROM order_items oi

LEFT JOIN products p

ON oi.product_id = p.product_id;



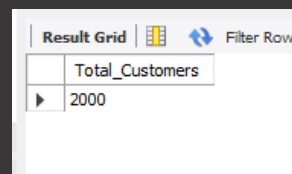
Total_Cost
5.839

-- Total Customers KPI

SELECT

COUNT(*) AS Total_Customers

FROM customers;



Total_Customers
2000

-- YTD Total Sales KPI

SELECT

year(o.order_date),

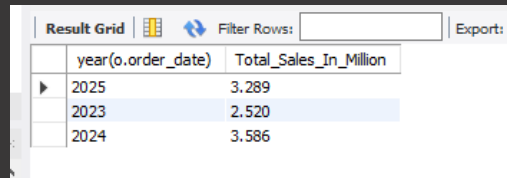
round(sum(oi.quantity * oi.unit_price)/1000000,3) AS Total_Sales_In_Million

```
FROM orders o
```

```
LEFT JOIN order_items oi
```

```
ON o.order_id = oi.order_id
```

```
GROUP BY YEAR(order_date);
```



year(o.order_date)	Total_Sales_In_Million
2025	3.289
2023	2.520
2024	3.586

-- MTD Total_Sales

```
SELECT
```

```
    MONTH(o.order_date) AS MTD_Total_Sales,
```

```
    YEAR(order_date) AS YTD,
```

```
    round(sum(oi.quantity *  
oi.unit_price)/1000,2) AS  
Total_Sales_In_Million
```

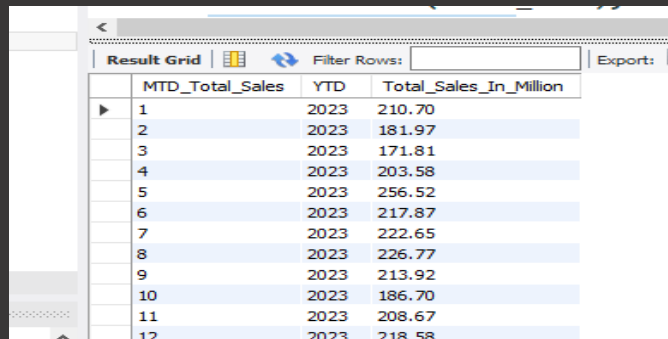
```
FROM orders o
```

```
LEFT JOIN order_items oi
```

```
ON o.order_id = oi.order_id
```

```
group by MTD_Total_Sales, ytd
```

```
order by ytd asc , mtd_total_sales asc;
```



	MTD_Total_Sales	YTD	Total_Sales_In_Million
1		2023	210.70
2		2023	181.97
3		2023	171.81
4		2023	203.58
5		2023	256.52
6		2023	217.87
7		2023	222.65
8		2023	226.77
9		2023	213.92
10		2023	186.70
11		2023	208.67
12		2023	218.58

-- Previous Year sales KPI

```
SELECT
```

```
    year(o.order_date) Years,
```

```
    round(sum(quantity * unit_price)/1000000,3) as  
YTD,
```

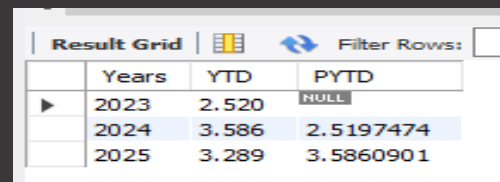
```
    lag(sum(quantity * unit_price )/1000000,1)  
over(order by year(o.order_date)) as PYTD
```

```
from orders o
```

```
left join order_items oi
```

```
on o.order_id = oi.order_id
```

```
group by year(order_date);
```

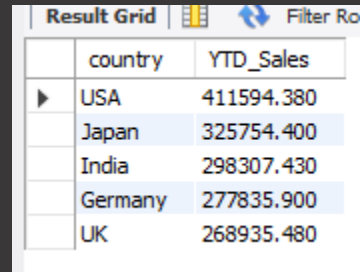


	Years	YTD	PYTD
	2023	2.520	NULL
	2024	3.586	2.5197474
	2025	3.289	3.5860901

----- Charts Requirements -----

-- TOP 5 Country Total Sales (IN 2023)

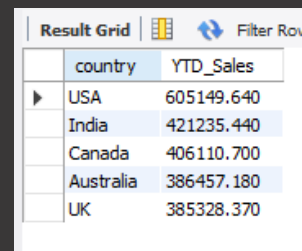
```
SELECT
    country,
    sum(oi.quantity * oi.unit_price) as YTD_Sales
From order_items oi
left join orders o
on oi.order_id= o.order_id
left join customers c
on o.customer_id = c.customer_id
Where year(Order_date) = 2023
GROUP BY country
ORDER BY sum(oi.quantity * oi.unit_price) DESC LIMIT 5;
```



	country	YTD_Sales
▶	USA	411594.380
	Japan	325754.400
	India	298307.430
	Germany	277835.900
	UK	268935.480

-- TOP 5 Country Total Sales (IN 2024)

```
SELECT
    country,
    sum(oi.quantity * oi.unit_price) as YTD_Sales
From order_items oi
left join orders o
on oi.order_id= o.order_id
left join customers c
on o.customer_id = c.customer_id
Where year(Order_date) = 2024
GROUP BY country
ORDER BY sum(oi.quantity * oi.unit_price) DESC LIMIT 5;
```



	country	YTD_Sales
▶	USA	605149.640
	India	421235.440
	Canada	406110.700
	Australia	386457.180
	UK	385328.370

-- TOP 5 Country Total Sales (IN 2025)

```
SELECT
    country,
    sum(quantity * unit_price) AS YTD_Sales
FROM
    order_items oi
```

```

LEFT JOIN orders o
on oi.order_id = o.order_id
LEFT JOIN customers c
ON o.customer_id = c.customer_id
WHERE YEAR(order_date) = 2025
GROUP BY Country
ORDER BY total_Sales DESC LIMIT 5;

```

	country	YTD_Sales
▶	USA	579602.850
	UK	380332.950
	Japan	368868.600
	India	356865.920
	Australia	354192.650

-- YTD Total Sales BY Channel (IN 2023)

```

SELECT
    channel,
    sum(quantity * unit_price) AS YTD_Sales
FROM orders o
LEFT JOIN order_items oi
ON o.order_id = oi.order_id
WHERE YEAR(order_date)=2023
GROUP BY channel;

```

	channel	YTD_Sales
▶	Web	1157645.560
	Mobile App	971587.750
	Marketplace	390514.060

-- YTD Total Sales BY Channel (IN 2024)

```

SELECT
    channel,
    sum(quantity * unit_price) AS YTD_Sales
FROM orders o
LEFT JOIN order_items oi
ON o.order_id = oi.order_id
WHERE YEAR(order_date)=2024
GROUP BY channel;

```

	channel	YTD_Sales
▶	Marketplace	544608.840
	Web	1601697.160
	Mobile App	1439784.090

-- YTD Total Sales BY Channel (IN 2025)

```

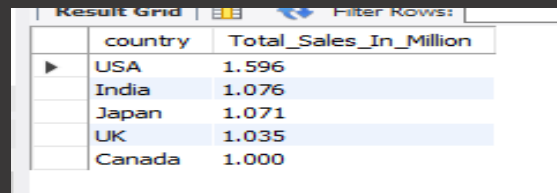
SELECT
    channel,
    sum(quantity * unit_price) AS YTD_Sales
FROM orders o
LEFT JOIN order_items oi
ON o.order_id = oi.order_id
WHERE YEAR(order_date)=2025
GROUP BY channel;

```

	channel	YTD_Sales
▶	Mobile App	1396972.590
	Marketplace	508627.980
	Web	1383734.450

-- TOP 5 Country BY Total Sales

```
SELECT
    c.country,
    round(SUM(oi.quantity * oi.unit_price)/1000000,3) as Total_Sales_In_Million
FROM customers c
LEFT JOIN orders o
on c.customer_id = o.customer_id
LEFT JOIN order_items oi
ON o.order_id = oi.order_id
GROUP BY c.country
ORDER BY SUM(oi.quantity * oi.unit_price)
DESC limit 5;
```

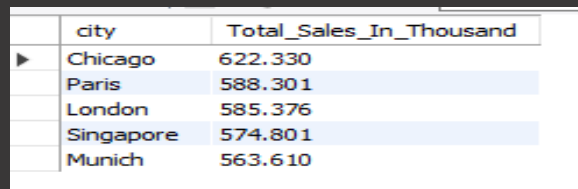


The screenshot shows a 'Result Grid' window with a table containing the top 5 countries by total sales. The table has two columns: 'country' and 'Total_Sales_In_Million'. The rows are: USA (1.596), India (1.076), Japan (1.071), UK (1.035), and Canada (1.000). The first row is highlighted with a blue background.

country	Total_Sales_In_Million
USA	1.596
India	1.076
Japan	1.071
UK	1.035
Canada	1.000

-- TOP 5 City BY Customers City

```
SELECT
    c.city,
    round(SUM(oi.quantity * oi.unit_price)/1000,3) as Total_Sales_In_Thousand
FROM customers c
LEFT JOIN orders o
on c.customer_id = o.customer_id
LEFT JOIN order_items oi
ON o.order_id = oi.order_id
GROUP BY c.city
ORDER BY SUM(oi.quantity * oi.unit_price) DESC limit 5;
```



The screenshot shows a 'Result Grid' window with a table containing the top 5 cities by total sales. The table has two columns: 'city' and 'Total_Sales_In_Thousand'. The rows are: Chicago (622.330), Paris (588.301), London (585.376), Singapore (574.801), and Munich (563.610). The first row is highlighted with a blue background.

city	Total_Sales_In_Thousand
Chicago	622.330
Paris	588.301
London	585.376
Singapore	574.801
Munich	563.610

-- TOP 5 Product BY Total Sales

```
SELECT
    p.product_name,
    round(SUM(oi.quantity * oi.unit_price)/1000,3) as Total_Sales_In_Thousand
FROM products p
```

```

LEFT JOIN order_items oi
ON p.product_id = oi.product_id
GROUP BY p.product_name
ORDER BY SUM(quantity * unit_price) DESC
limit 5;

```

product_name	Total_Sales_In_Thousand
Pro Monitor	176.332
Eco Keyboard	175.771
Ultra Headphones	160.538
Pro Charger	148.407
Classic Tablet	120.337

-- TOP 5 Category BY Total_Sales

```

SELECT
    p.category,
    round(SUM(oi.quantity * oi.unit_price)/1000,3) as Total_Sales_In_Thousand
FROM products p
LEFT JOIN order_items oi
ON p.product_id = oi.product_id
GROUP BY p.category
ORDER BY SUM(quantity * unit_price) DESC limit

```

category	Total_Sales_In_Thousand
Electronics	3921.729
Home & Kitchen	1559.646
Sports & Outdoors	1212.176
Toys & Games	824.334
Clothing	806.943

5;

-- TOP 5 Country BY Profit

```

SELECT
    country,
    SUM(quantity * unit_price)- SUM(quantity *Cost_Price) as Total_Profit
FROM order_items oi
LEFT JOIN products p
on oi.product_id = p.product_id
LEFT JOIN orders o
ON oi.order_id = o.order_id
LEFT JOIN customers c
ON c.customer_id = o.customer_id
GROUP BY Country
ORDER BY Total_Profit DESC LIMIT 5;

```

country	Total_Profit
USA	608684.380
Japan	407796.190
India	407079.040
UK	392808.940
Canada	375416.410

-- Total_Profit BY Channel

```

SELECT
    channel,
    SUM(quantity * unit_price)- SUM(quantity *Cost_Price) as Total_Profit
FROM order_items oi
LEFT JOIN products p
on oi.product_id = p.product_id
LEFT JOIN orders o
ON oi.order_id = o.order_id
GROUP BY channel

```

channel	Total_Profit
Web	1579429.340
Mobile App	1436109.940
Marketplace	540323.540

```
ORDER BY Total_Profit DESC LIMIT 5;
```

-- TOP 5 CITY BY Profit

```
SELECT
```

```
    c.city,
```

```
    SUM(oi.quantity * oi.unit_price)- SUM(oi.quantity *p.Cost_Price) as Total_Profit
```

```
FROM order_items oi
```

```
LEFT JOIN products p
```

```
on oi.product_id = p.product_id
```

```
LEFT JOIN orders o
```

```
ON oi.order_id = o.order_id
```

```
LEFT JOIN customers c
```

```
ON c.customer_id = o.customer_id
```

```
GROUP BY c.city
```

```
ORDER BY Total_Profit DESC LIMIT 5;
```

	city	Total_Profit
▶	Chicago	234401.000
	London	222831.030
	Paris	220855.200
	Singapore	218517.870
	Tokyo	216593.680

-- TOP 5 Product BY Profit

```
SELECT
```

```
    p.product_name,
```

```
    SUM(oi.quantity * oi.unit_price)- SUM(oi.quantity *p.Cost_Price) as Total_Profit
```

```
FROM order_items oi
```

```
LEFT JOIN products p
```

```
on oi.product_id = p.product_id
```

```
LEFT JOIN orders o
```

```
ON oi.order_id = o.order_id
```

```
LEFT JOIN customers c
```

```
ON c.customer_id = o.customer_id
```

```
GROUP BY p.product_name
```

```
ORDER BY Total_Profit DESC LIMIT 5;
```

	product_name	Total_Profit
▶	Eco Keyboard	73289.490
	Pro Monitor	61294.840
	Ultra Headphones	55981.570
	Premium Tablet	53020.490
	Pro Charger	52527.740

Project Developed By : Ghulam Mujtaba